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Hardware-Accelerated Digital Power Control for High-Frequency Hybrid Energy Storage Systems Using MCUs

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ABSTRACT

In the rapidly evolving field of electric vehicles (EVs), efficient energy storage systems are crucial for widespread adoption. Hybrid energy storage systems (HESS), which combine lithium batteries with supercapacitors (SCs), offer a promising solution by improving power density and overall system efficiency. This paper presents a cost-effective approach to implementing high-frequency current controllers within an HESS using the general-purpose microcontroller STM32G474RB. By leveraging its built-in filter math accelerator (FMAC), a type II compensator is implemented, achieving 250 kHz current control and 500 kHz switching frequency. This enhances computational efficiency by 33% compared to using only the central processing unit (CPU) for calculations. This approach reduces system size and cost, providing a viable alternative to more expensive digital signal processor (DSP) and field-programmable gate array (FPGA) solutions. The proposed design is validated through hardware implementation, demonstrating its potential for enhancing HESS performance in EVs.

1 | Introduction

In the rapidly evolving field of electric vehicles (EVs), one of the most pressing challenges is the development of energy storage systems [1, 2]. This issue significantly hampers the advancement and widespread adoption of EVs [3, 4]. A promising solution is the implementation of hybrid energy storage systems (HESS), which combine lithium batteries with supercapacitors (SCs). This well-designed HESS enhances the power density of the storage system and increases overall system efficiency [5, 6]. By absorbing regenerative braking energy through SCs, it can also extend the lifespan of lithium batteries.

A semi-active HESS consists of a single DC-DC converter to control the current of the SC. It is widely used because it achieves an optimal balance between economy and performance [7–10]. However, since the semi-active architecture cannot directly control the direction of the battery current, it has higher requirements on the performance of the controller. In response to the continuous advancements in EV performance, there is an increasing demand for higher power DC-DC converters. However, higher power typically results in larger size inductors. To maintain the compactness of the HESS, it is essential to increase the switching frequency of these converters. Currently, most switch mode power sources operate at frequencies between 10 kHz and 100 kHz [11–19]. Achieving a higher switching frequency necessitates an elevated control frequency, requiring more higher computational performance and compact designs.

Table 1 shows different digital power schemes in recent research. Traditionally, field-programmable gate arrays (FPGAs) and digital signal processors (DSPs) have been the go-to solutions for

TABLE 1 | Comparison of different digital power schemes.

Converter topology	Processing unit	Controller	Current/voltage control frequency	Switch frequency
Boost [11]	dSPACE	PI	–	20 kHz
Buck [13]	dSPACE	Constant power generation	–	20 kHz
Buck [12]	DSP	PI	–	10 kHz
Buck [15]	DSP	Model predictive control	1 kHz	10 kHz
Boost [17]	DSP	Model predictive control	1 kHz	20 kHz
Buck [18]	FPGA	Local model-based control	–	10 kHz
Buck [20]	FPGA	One pole one zero compensator	3 MHz	3 MHz
Power factor correction [22]	MCU + CPLD	PI	–	400 kHz
Full bridge DCDC [19]	DSP + FPGA	–	–	10 kHz

high-frequency digital switch mode power supply control [12, 13, 15, 17, 18, 20]. FPGAs excel at implementing control algorithms and achieving high control frequencies due to their hardware computation capabilities. However, they often lack integrated analog-to-digital converters (ADCs), requiring additional modules that increase system size and potentially introduce delays [16, 18, 20]. DSPs are more specialized for digital power applications. They feature built-in high-speed ADCs and communication peripherals, facilitating the implementation of control algorithms using standard programming languages. Some DSPs, such as the C2000 series, include a control law accelerator (CLA)—a specialized coprocessor that operates in parallel with the main CPU, enhancing control frequency and overall efficiency [21]. Despite these advantages, both DSPs and FPGAs are generally more expensive than general-purpose microcontrollers (MCUs).

The use of multiple coprocessors for control has gained popularity in recent years. In addition to employing DSPs and CLA as coprocessors, the combination of MCUs with programmable logic devices (PLDs) has also been adopted. In [22], the author utilizes an MCU for ADC sampling, pulse-width modulated (PWM) waveform generation, and proportional-integral (PI) controller computation. Additionally, due to the extensive use of discrete logic gates for selecting control signal outputs, a complex programmable logic device (CPLD) is employed alongside the MCU for system control. This can reduce software judgment logic in MCU to improve computational performance. The use of DSP with FPGA is also adopted in [19]. It presents the development board for EV applications, integrating both a DSP and an FPGA. In practical use, the DSP serves as the master controller, while the FPGA acts as a coprocessor, handling computationally intensive control algorithms at high frequencies and returning the results to the DSP. This approach has the potential to achieve higher switching frequencies. However, it comes with higher costs and requires a larger size due to the DSP and FPGA integration.

This paper demonstrates the use of a general-purpose MCU, such as the STM32G4 series, as an affordable alternative to more expensive DSPs and FPGAs. Recent advancements in semiconductor technology have significantly enhanced MCU capabilities, including the integration of specialized features like the filter math accelerator (FMAC). This feature enables the MCU to perform digital filtering or compensation tasks as an internal coprocessor, removing the need for additional FPGAs or CPLDs, which helps reduce both cost and size. The study focuses on implementing a type II compensator using the built-in FMAC of the STM32G474RB MCU, achieving a current control frequency of 250 kHz. This demonstrates that high-frequency current controllers can be deployed without relying on costly DSPs or FPGAs. The proposed approach not only reduces system costs but also preserves high computational performance. By employing general-purpose MCUs for high-frequency current control, this study advances EV energy storage, offering a cost-effective solution that supports compact HESS designs and improved EV performance. Additionally, the approach holds potential for applications in renewable energy, industrial automation, and consumer electronics.

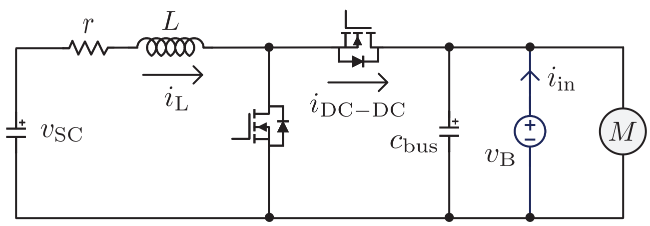

FIGURE 1 | 2-switch buck-boost circuit.

This rest of this article is organized as follows. Section 2 introduces the system architecture, including the control objectives and the energy management algorithm. Section 3 presents the hardware circuit design, covering current and voltage sampling and the gate drive circuit. Section 4 discusses the configuration of MCU peripherals and their advantages. Section 5 explains the deployment of the FMAC. Section 6 presents and discusses the experimental results. Section 7 concludes this article, and analyses the problems with the current scheme and possible improvements.

2 | System Structure

As illustrated in Figure 1, the converter employs a two-switch buck-boost topology, enabling bidirectional energy transfer. The

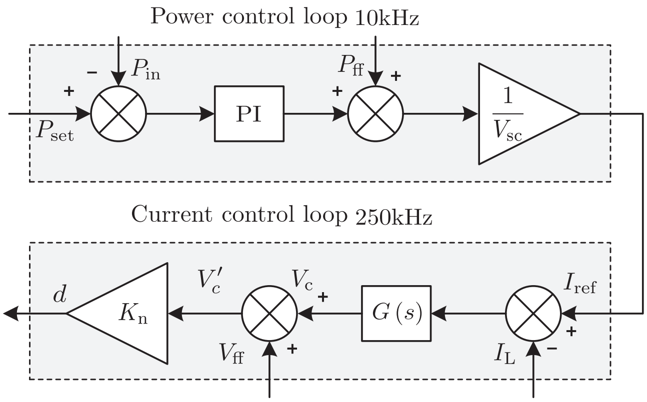

FIGURE 2 | HESS control structure.

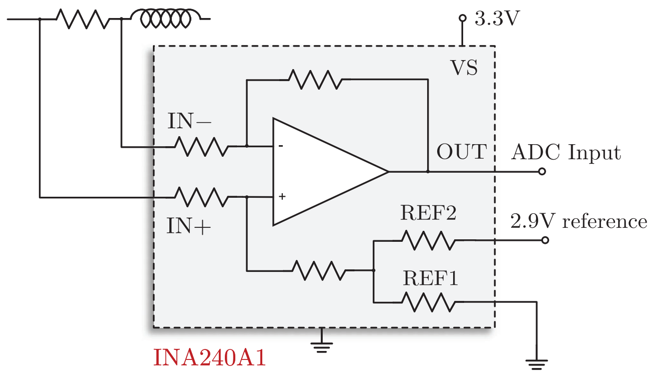

FIGURE 3 | Current sampling circuit.

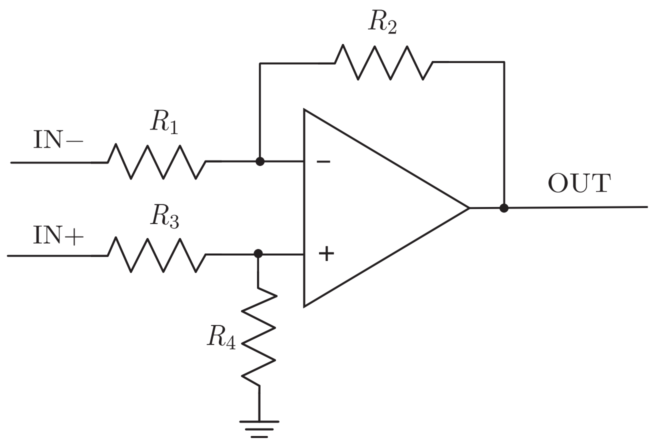

FIGURE 4 | Differential amplifier circuit.

converter operates in synchronous rectification mode. When applied in a HESS, the maximum voltage of the SC must be maintained below the electrical bus voltage. This ensures that when energy flows from the SC to the electrical bus, the converter functions in boost mode, with the lower switch serving as the main metal-oxide-semiconductor field-effect transistor (MOSFET) and the upper switch as the synchronous rectification MOSFET. Conversely, when energy flows from the electrical bus to the SC, the converter operates in buck mode, where the upper switch acts as the main MOSFET and the lower switch functions as the synchronous rectification MOSFET. This design follows a semi-active HESS architecture, where the battery output power, P_{in}, is indirectly regulated by controlling the output power of the DC-DC converter, $P_{\mathrm{DC-DC}}$. Assuming negligible circuit losses, the system power balance can be expressed as:

$$P_{\mathrm{in}} + P_{\mathrm{DC-DC}} = P_{\mathrm{out}} \tag{1}$$

where P_{out} represents the motor power. From this equation, when the motor power or load power remains constant, adjusting the $P_{\mathrm{DC-DC}}$ will influence P_{in}. In circuit terms, due to the presence of wiring impedance and the internal resistance of the battery, the battery and the DC-DC converter share the load in a manner similar to parallel branches in an electrical circuit. The feedforward power output P_{ff} from the DC-DC can be obtained from (1).

$$P_{\mathrm{ff}} = P_{\mathrm{out}} - P_{\mathrm{set}} \tag{2}$$

where P_{set} is the battery reference power. Since wiring losses and other factors were initially neglected in the modeling process, the sum of the battery and converter power does not perfectly match the load power. Consequently, relying solely on feedforward control results in a steady-state error. To address this, a PI controller is introduced as the first closed-loop in Figure 2. The power control loop operates at 10 kHz, determining the required SC power, which is then divided by the SC voltage to generate the reference current for the current loop. This current control loop, incorporating a serial compensator and a feedforward component, is computed at 250 kHz. In buck mode, the feedforward voltage V_{ff} is SC voltage V_{SC}, and in boost mode, it is electrical bus voltage V_{B}. Finally, the output voltage V_c' is normalized through division by the battery voltage to obtain the duty cycle d. The detailed implementation will be discussed in Section 5.3 and Appendix A.

3 | Hardware Design

This article uses the factory-calibrated, high-precision 2.9 V voltage reference source of the STM32G4 as the reference for the ADC.

3.1 | Current and Voltage Sample

The MCU needs to use ADC sampling to get the current data on the inductor. The sampling resistor method is used for current sampling. The inductor current I_L sampling use INA240A1 with a gain of 20 (A_i) and a sampling resistor of 5mΩ (R_s). As shown in Figure 3, by grounding the REF1 pin and connecting the REF2 pin to a 2.9 V reference, the output voltage will have a offset voltage V_{offset} that equal to half of 2.9 V reference.

$$V_{\mathrm{out}} = I_L \times R_s \times A_i + V_{\mathrm{offset}} \tag{3}$$

In Figure 4, a differential amplifier is chosen for voltage sampling to improve the signal-to-noise ratio, as it effectively eliminates common-mode noise in voltage measurements. When $\mathrm{R}_1 = \mathrm{R}_3 = 10\,\mathrm{k}\Omega$, $\mathrm{R}_2 = \mathrm{R}_4 = 1\,\mathrm{k}\Omega$, the voltage sampling gain A_v is 10.

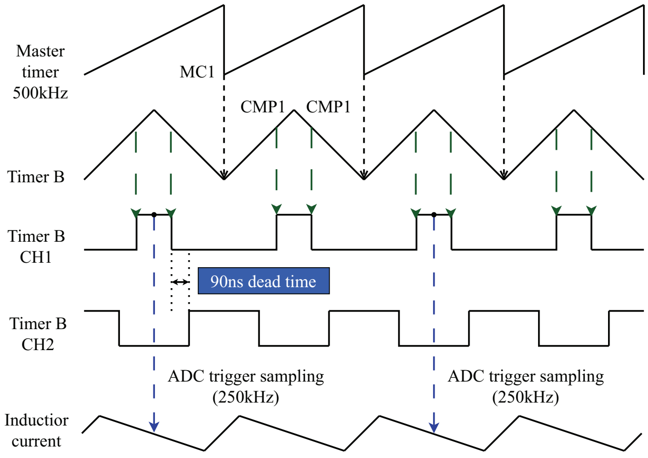

FIGURE 5 | HRTIM sampling and output timing chart.

3.2 | Gate Driver

UCC27211 is a 4 A, 120 V half bridge gate driver with transistor–transistor logic inputs. It can be driven by the 3.3 V PWM signal output from the MCU. The upper switch is driven using a 100 nF bootstrap capacitor configuration. Due to the presence of capacitor leakage current, it cannot maintain conduction for a long time and needs to be turned off to charge the bootstrap capacitor. Therefore, the upper MOSFET in this circuit cannot achieve a 100% duty cycle. In this circuit, the duty cycle is limited to 94%. The switching frequency is 500 kHz.

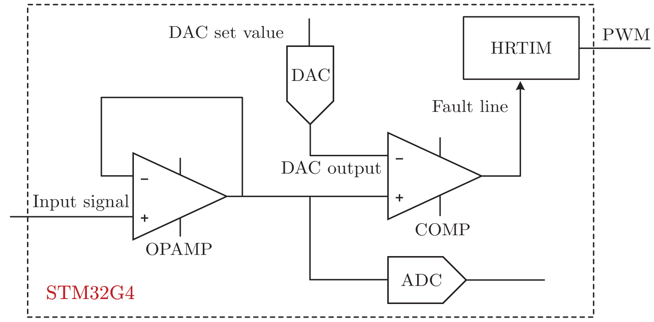

FIGURE 6 | Overcurrent and overvoltage protection functional block diagram.

TABLE 2 | Comparison of different protection methods with colored advantages and disadvantages.

Feature	Software protection	External hardware protection	Chip peripheral protection
Cost	Low	High	Low
Integration	High	Low	High
Threshold changeability	Easy	Difficult	Easy
Response speed	Slow	Fast	Fast
CPU load	High	Low	Low

4 | MCU Peripherals Configuration

The MCU used for the controller is the STM32G474RB with a clock frequency of 170 MHz, and to achieve a current loop control frequency of up to 250 kHz, the code needs to be extremely optimized and accelerated by FMAC for hardware acceleration of the compensator. The software running on MCU consists of two main control loops, a 10 kHz power loop, scheduled by timer interrupts, and a 250 kHz current loop, scheduled by direct memory access (DMA) and FMAC interrupts. One-time current loop calculation requires two times CPU calls and uses three peripheral modules: ADC, FMAC, and high-resolution timer (HRTIM).

According to Figure 5, timer B is triggered by the master timer to perform counting. To simplify mean inductor current sampling, configure timer B in up–down counting mode. In this mode, the counter counts up to the set value, then counts back down, creating a PWM pulse that is symmetrical at the pulse midpoint. This midpoint is ideal for sampling voltage and current data due to its lower noise. As a result, this method reduces measurement noise, harmonic distortion, and electromagnetic interference.

4.1 | Overcurrent and Overvoltage Protection Circuit

The MCU leverages its internal peripherals to achieve hardware-based over-voltage and over-current protection. This integrated approach utilizes the built-in digital-to-analogue converters (DACs), comparators, and operational amplifiers.

As shown in Figure 6, the voltage signal is first fed into an internal voltage follower to increase the input signal impedance. The output from the voltage follower is sampled by the ADC and also fed into the positive input of a comparator. The negative input of the comparator is connected to the output of the DAC, which can adjust its output voltage based on the protection trigger threshold. When the input voltage signal exceeds its output, the comparator outputs a high level to the fault line. To avoid system mis-triggering due to noise, the fault line is equipped with a filter that can be set to trigger a fault only if multiple consecutive PWM cycles are high. In case of an error, an external event triggers the HRTIM, which promptly shuts down the PWM output, ensuring system protection. This design fully leverages the internal peripheral resources of the MCU. As shown in Table 2, this approach combines the advantages of both software and hardware protection, providing a cost-effective and highly integrated solution for overcurrent and overvoltage protection.

4.2 | Calculate Sequence

The key to high-frequency current loop control lies in the synchronous computation enabled by the FMAC. As shown in Figure 7, the ADC begins the sampling process upon receiving a trigger from the HRTIM. After sampling is complete, the ADC generates a DMA request. The DMA then transfers the data from

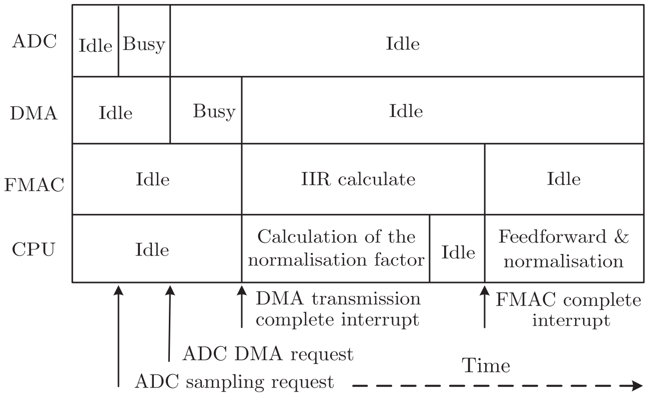

FIGURE 7 | Current loop calculation sequence diagram.

ADC to CPU memory and FMAC register. The DMA will generate an interrupt after completing the data transfer, and the CPU will calculate the normalization factor in this interrupt function. It will generate an interrupt after completing the data transfer, and the CPU will calculate the normalization factor in this interrupt function. At the same time, when new data enters FMAC, the FMAC will begin infinite impulse response (IIR) calculations. The inclusion of DMA also prevents the CPU from being involved in data transfer, thereby avoiding excessive CPU load.

Typically, FMAC computations take longer than the CPU normalization factor calculations. After the CPU completes its task, it enters an idle state, during which it can execute lower-priority tasks. As shown in Figure 8, by assigning higher priority to interrupts with strict real-time requirements, the system ensures that high-priority interrupts can preempt lower-priority ones when current loop computations are needed. Since the interrupt priority for FMAC completion is lower than that for DMA completion, the FMAC completion interrupt can only be triggered once the DMA completion interrupt has finished. This configuration ensures the stability of the current loop computation timing.

The peripheral function block diagram for the current loop calculation flow is shown in Figure 9. It provides a more detailed illustration of the interrupt calls and data flow between peripherals. The whole current loop timing, through the application of DMA, reasonable interrupt triggering, and priority arrangement, achieves ensuring the high real-time current loop control calculation while reducing the CPU load and the CPU current loop calculation idle time to perform other calculation tasks.

5 | FMAC Type II Compensator Implement

5.1 | Type II Compensator Design

The current compensator is based on a type II compensator. In the z domain, it needs to use a 2$^{\text{nd}}$ IIR filter to deploy, which has two poles and three zeros. After discretizing the compensator using the bilinear method, the transfer equation of the type II compensator can be obtained.

$$H(z) = \frac{y(z)}{x(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{a_0 + a_1 z^{-1} + a_2 z^{-2}} \tag{4}$$

Reduced to the form of a differential expression and inverse Laplace transform

$$\begin{aligned} a_0 y(z) = b_0 x(z) + b_1 x[n-1] + b_2 x[n-2] \\ - a_1 y[n-1] - a_2 y[n-2] \end{aligned} \tag{5}$$

The transfer function for a type II compensator in the s domain is as follows.

$$G(s) = \frac{f_{p0}}{f_z} \frac{1 + \frac{f_z}{s}}{1 + \frac{s}{f_p}} \tag{6}$$

The values of the zero and pole parameters are as follows: $f_p =$ 269.98 kHz, $f_z = 2.1872$ kHz, and $f_{p0} = 8.2463$ kHz. According to Figure 10, after adding the compensator, the low-frequency gain of system significantly increases compared to the uncompensated case, which helps reduce steady-state error. In the mid-frequency range, the compensator maintains a higher gain and lifts the amplitude curve, thereby increasing the crossover frequency to 25 kHz. Near the crossover frequency, the compensated curve crosses 0 dB with a gentler slope, resulting in significantly improved system stability. Meanwhile, in the high-frequency range, the compensator suppresses the gain, helping to reduce high-frequency noise and interference. Overall, although the phase margin decreases from 90° to 80° after adding the compensator, it still effectively enhances low-frequency gain, increase crossover frequency and suppresses high-frequency gain, leading to improved steady-state accuracy and better dynamic performance of the system.

5.2 | Using ADC Offset Instead of Negative Feedback Subtraction

Referring to Figure 2, the input of the compensator is a deviation between the reference and actual currents. Traditional digital power supplies necessitate a CPU intervention to compute this discrepancy.

$$I_{\mathrm{e}} = I_{\mathrm{ref}} - I_{\mathrm{L}} \tag{7}$$

where I_{e} is negative feedback error current and I_{ref} is reference current. In this paper, the ADC offset feature is utilized to avoid CPU processing for subtraction tasks, as shown in Figure 11. This approach allows the ADC to directly subtract a predefined value N_{ref} from its sampled reading N_{L}, streamlining the error calculation process. At this time, the input of FMAC becomes the difference between two ADC values $-N_{\mathrm{e}}$, while the negative sign reverses.

$$-N_{\mathrm{e}} = N_{\mathrm{L}} - N_{\mathrm{ref}} \tag{8}$$

Therefore, after obtaining all parameters, the final step requires taking the negative of parameter b to compensate for the sign. For simplicity in subsequent derivations, assume the input FMAC variable is already N_e. In a 12-bit ADC, the offset value is

$$N_{\mathrm{ref}} = 2048 + \frac{4095 I_{\mathrm{ref}} R_s A_i}{V_{\mathrm{ref}}} \tag{9}$$

where V_{ref} is the ADC reference voltage.

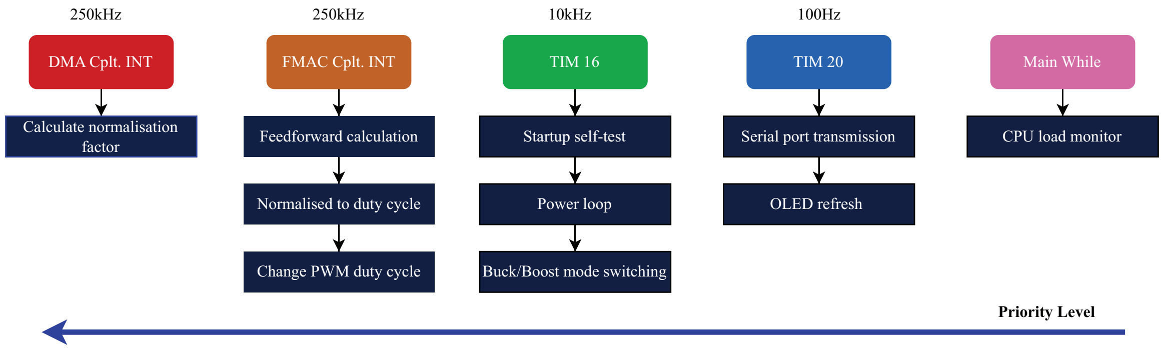

FIGURE 8 | MCU interrupt (INT) arrangement.

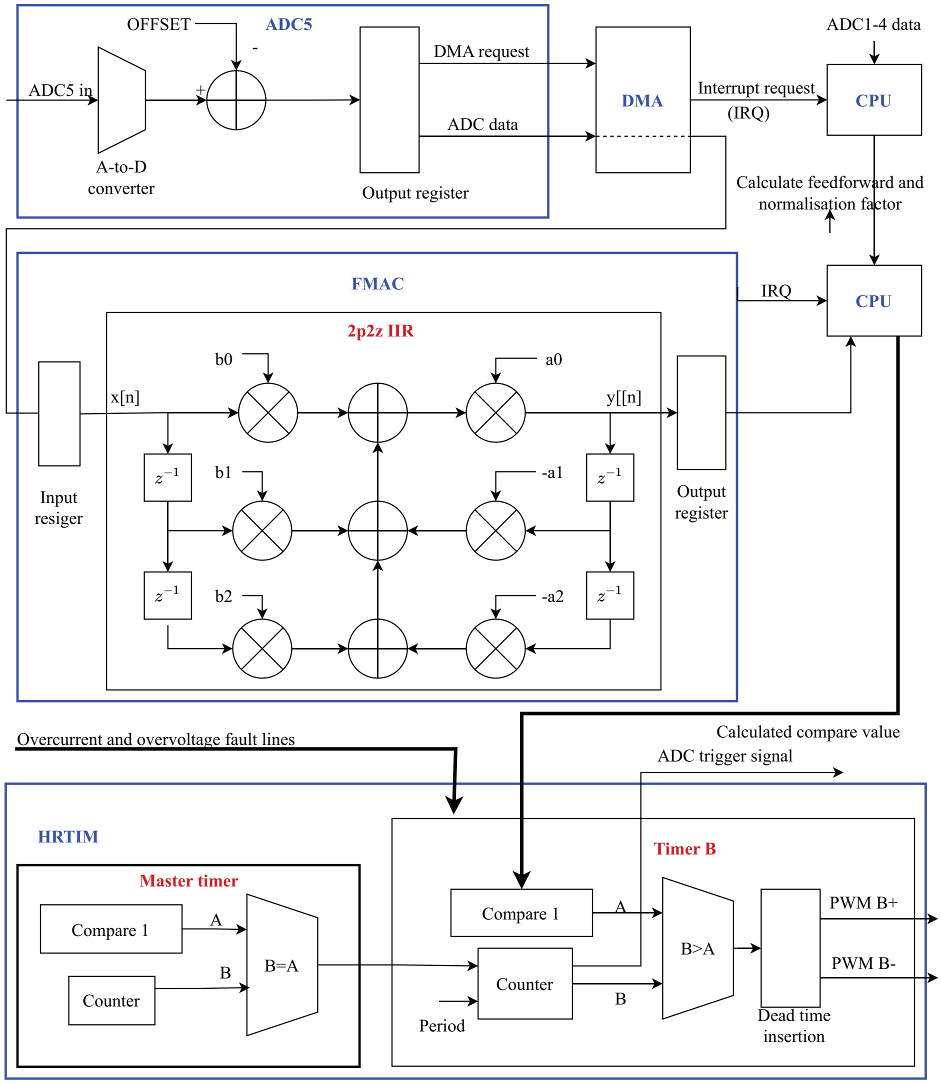

FIGURE 9 | Current loop peripheral function block diagram.

5.3 | Gain Compensation

Since the input to the designed compensator is the error current in Figure 2, but the input of FMAC is an ADC sample error signal N_e as shown in Figure 9, the gain of the compensator needs to be compensated. At the same time, the duty cycle formula based on the voltage calculation can be further simplified and converted into an expression based on the ADC sampling value to reduce the CPU operation. After simplification, the duty cycle formula is as follows. The simplification process can be found in Appendix A.

$$d = \frac{1}{N_B}\left(N_e \cdot \frac{1}{A_v A_i R_s} G + N_{\mathrm{ff}}\right) \tag{10}$$

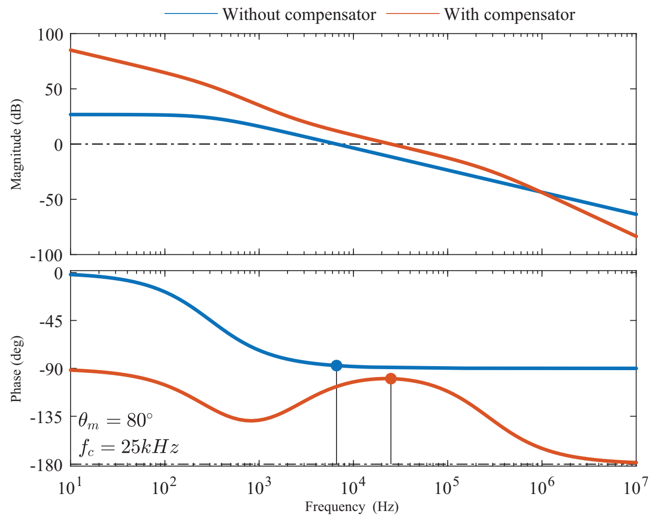

FIGURE 10 | Bode plot of current control open loop.

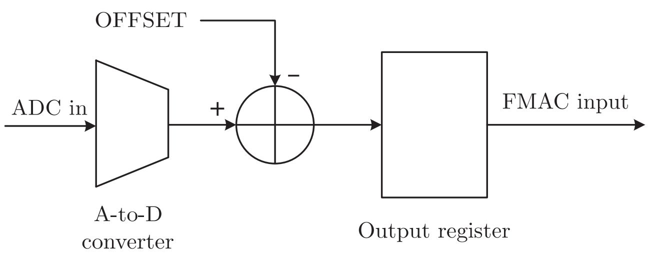

FIGURE 11 | ADC function diagram.

where N_B represents the ADC sampling value, and N_{ff} is the feedforward voltage value. In buck mode, it corresponds to the SC voltage sampling value, while in boost mode, it represents the sampling value of the bus voltage. The final result is normalized by dividing by N_B. The term $Ne \cdot \frac{1}{A_v A_i R_s} G$ refers to the output value of the FMAC where $1/(A_v R_i R_s)$ represents the compensation gain of the compensator.

With this simplification, it is possible to convert most of the multiplication calculations of floating-point numbers into integers, greatly saving CPU time. After obtaining the FMAC output, only one addition $(+N_{ff})$ and one multiplication $(\times \frac{1}{N_B})$ are required to complete the feedforward and normalization.

6 | Experimental Results

The experimental platform of this SMPS is shown in Figure 12. In the experimental setup, the digital power supply operates in constant voltage mode, serving as a substitute for the battery. As shown in Figure 13, the HESS controller is a hardware module that includes converters, MCU, and auxiliary power supply, and it operates without the need for any additional power supply. Its dimensions are 6.5×5.5 cm. The design allows for a dual converter interleaved parallel configuration, and in single converter mode, its continuous current can reach up to 7 A.

To monitor CPU workload, when the CPU is idle, it continuously inverts the output of a specific pin. By observing the voltage

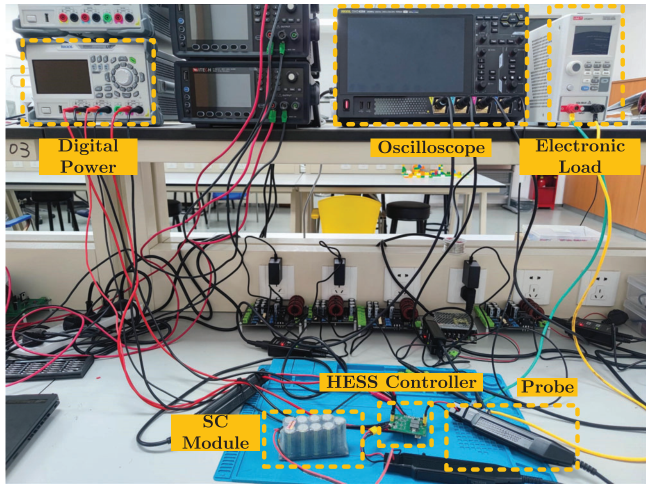

FIGURE 12 | Experimental platform.

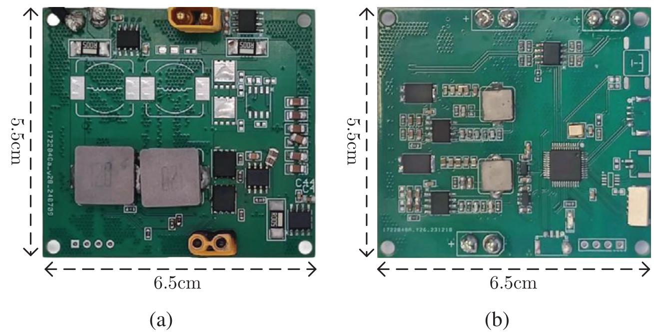

FIGURE 13 | Printed circuit board of HESS controller. (a) Top layer. (b) Bottom layer.

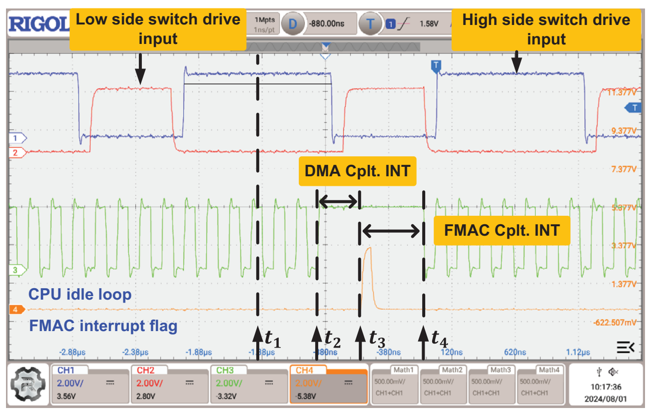

FIGURE 14 | CPU load monitor.

changes at this pin, the occupancy time of the CPU under different timing conditions can be measured. When the CPU is idle, the voltage level of pin can be observed toggling continuously on the oscilloscope. When the CPU is busy, the voltage level will remain steady. To indicate the start of the FMAC interrupt, the CPU inverts the voltage on another pin twice at the beginning of the FMAC interrupt function.

Figure 14 illustrates that at time t_2, both the CPU and FMAC receive the latest data sampled by the ADC and begin

TABLE 3 | Time consumed for one time closed-loop computation.

	ADC	DMA	FMAC	CPU
Time (ns)	352	148	400	800
Improved computing efficiency		33%		

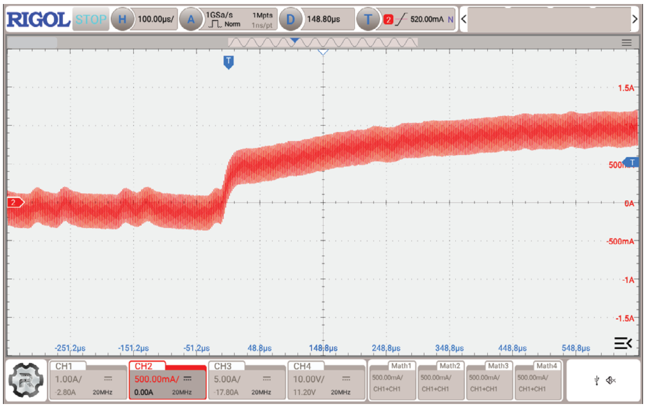

FIGURE 15 | Step response.

computation. Time t_1 marks the start of ADC sampling, which occurs at the midpoint of the high-side switch on-time. The interval between t_1 and t_2 is 500 ns, covering the total time for ADC sampling and DMA data transfer, during which the CPU remains idle. The ADC frequency of the STM32G4 is 42.5 MHz, requiring 15 clock cycles to complete sampling, resulting in an ADC sampling time of 352 ns. The ADC sampling consumes a significant portion of the time.

$$t_s = 15 \times \frac{1}{42.5M} = 352 \text{ ns} \tag{11}$$

At time t_2, the FMAC and CPU start working simultaneously. After completing the feedforward computation in the DMA interrupt, the CPU immediately executes the FMAC interrupt function at time t_3. There is no CPU idle in two interrupts as shown in Figure 7. This sequence indicates that the FMAC completes its computation earlier than the CPU. However, because the DMA interrupt has a higher priority than the FMAC interrupt, the CPU only executes the subsequent FMAC interrupt computations after completing the feedforward calculation. The computation timing is rigorously maintained through careful interrupt priority configuration.

The entire process occupies 800 ns of CPU time, and at a control frequency of 250 kHz, the current loop only utilizes 20% of its processing capacity. The addition of FMAC parallel computing reduces the computation time by up to 400 ns, which is a 33% reduction compared to the final 800 ns. Table 3 better illustrates the time spent in each stage.

As illustrated in Figure 15, the current controller achieves a fast step response with a settling time of 400 µs, much shorter than the 2 ms of the model predictive current controller in [15]. Both controllers have zero steady-state error and no overshoot, showing that the proposed controller performs similarly to mainstream control methods while offering a low-cost solution for HESS systems.

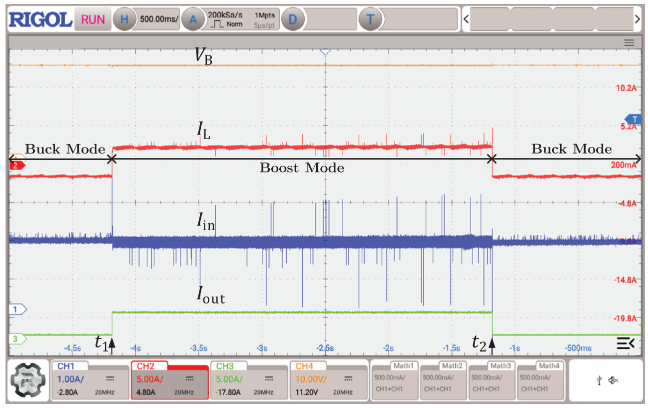

FIGURE 16 | Load step response.

The primary issue with general-purpose MCU is the relatively low frequency of the ADC. With a sampling time of 352 ns at a switching frequency of 500 kHz, the ADC is already struggling to keep up. This could lead to situations where a single sampling period spans two distinct switching states, potentially introducing some error in the sampled current. Moreover, to improve accuracy, it is necessary to compensate for the sampling time when triggering center-aligned sampling. By triggering the sampling slightly earlier, the accuracy of the results can be further enhanced.

Figure 16 illustrates the operating conditions of the HESS system under a step change of the electronic load. At time t_1, the load current increases from 0.5 A to 2.5 A, prompting the DC-DC converter to switch between boost and buck modes, and then returns to 0.5 A at t_2 to maintain a constant input current. This rapid power response is primarily enabled by the feedforward control of the 10 kHz power loop, which swiftly adjusts the reference of the inner current loop when a load disturbance occurs. Consequently, the inner loop rapidly modulates the target power value, ensuring that the battery output current remains nearly constant. This demonstrates that the low-cost HESS can maintain stable battery output and respond rapidly to varying power demands.

As illustrated in Figure 17, the deployment on the simulation EV platform enables the three driving processes of acceleration, cruising, and braking for electric vehicles. Throughout these processes, the energy management algorithm ensures that the battery output power is consistently maintained at 50 W. The motor may require up to 100 W of power during acceleration, supplemented by SC discharging to meet the additional demand. In the cruising stage, the motors consume approximately 80 W, with the SC discharging to fulfill this requirement. A reverse current is generated during braking, and the SC completely absorbs the recovered energy. The battery output remains stable at 50 W throughout, with no recovered energy being charged. Consequently, the HESS achieves the intended performance.

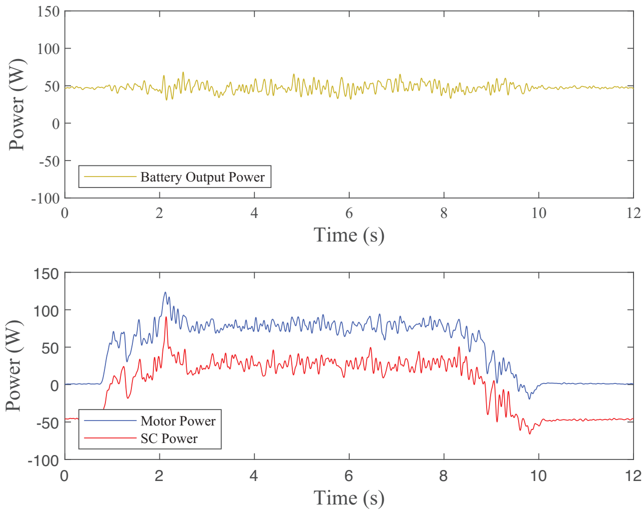

FIGURE 17 | EV deployment test.

Currently, most digital controllers implemented using DSPs have not exceeded the control frequency achieved in this study [12, 13, 15, 16]. In [20], a 400 MHz FPGA was used to implement a cycle-by-cycle controlled buck converter. The system completes one control computation in 80 FPGA clock cycles, which requires a total computation time of 200 ns, and it achieves a maximum control and switching frequency of 3 MHz. Although the computation time of MCU is 800 ns, which is longer than that of the FPGA, it is important to consider that the clock frequency of FPGA is 2.35 times higher than the MCU (170 MHz) and the Artix-7 FPGA costs approximately $30, which is ten times the cost of the $3 MCU. Moreover, the FPGA has a larger chip package and requires external ADCs. Therefore, FPGAs should not be regarded as the only option for achieving high frequencies in the range of 200 kHz to 1 MHz, and MCUs are clearly more suitable in terms of both cost and packaging.

7 | Conclusion

This paper presents a 2-switch bidirectional buck boost circuit for HESS using the built-in FMAC module in MCU. By employing parallel computation between the CPU and FMAC, a control frequency of 250 kHz is achieved. The proposed design optimizes the utilization of peripheral resources within the MCU, executing critical tasks such as overcurrent and overvoltage protection, negative feedback subtraction, partial normalization calculations, and synchronous voltage and current sampling. The implementation demonstrates exceptional computational efficiency, with the complete current loop calculation executed in a mere 800 ns. The incorporation of the FMAC module yields a significant 33% reduction in computation time while maintaining rapid control performance. This methodology establishes the feasibility of controlling high-frequency DC-DC converters in HESS applications using an MCU-based approach.

However, the current limitations in ADC frequency present a bottleneck for the entire system, constraining further increases in control frequency. Using an external high-speed ADC can significantly reduce ADC sampling delay. In [20], the LTC2378 external ADC reduces the sampling delay by half to 200 ns, meaning the system has the potential to achieve a control frequency of up to 500 kHz. High-speed digital control requires a fast ADC to prevent aliasing of the switching ripple in the sampled output voltage. However, since harmonic frequencies are integer multiples of the switching frequency, aliasing can be avoided by placing a notch filter before the ADC sampling stage, eliminating the need for a high-speed ADC [23, 24]. Future research directions could explore advancements in ADC sampling time to mitigate sampling delays or investigate alternative methodologies to ameliorate the impact of extended sampling periods on system performance.

Author Contributions

Wenhao Lin: formal analysis, investigation, methodology, writing – original draft. **Guanying Chu:** formal analysis, funding acquisition, methodology, supervision, writing – review & editing.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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Appendix A: Gain Calculation and Formula Simplification

The output of the compensator in the block diagram is,

$$v_c = I_e \cdot G. \tag{A1}$$

According to the current sample circuit in Section 3.1,

$$I_e = N_e \cdot \frac{V_{ref}}{4095 A_i R_s}. \tag{A2}$$

The compensator output requires gain adjustment to achieve the designed output controller voltage V_c.

$$V_c = N_e \cdot \frac{V_{ref}}{4095 A_i R_s} G \tag{A3}$$

Therefore, to compensate for the gain of the ADC and sampling circuit, the compensator output needs to be multiplied by K_G.

$$K_G = \frac{V_{ref}}{4095 A_i R_s} \tag{A4}$$

After multiplying, the output of FMAC is v_c which is the same as the compensator output in the block diagram (Figure 2). To reduce computational load, derive the duty cycle equation expressed in terms of ADC values and simplify it.

$$d = K_N (N_e \cdot K_G G + V_{\mathrm{ff}}) \tag{A5}$$

$$V_{\mathrm{ff}} = A_v N_{\mathrm{ff}} \cdot \frac{V_{\mathrm{ref}}}{4095} \tag{A6}$$

$$K_N = \frac{1}{V_B} = \frac{1}{A_v N_B} \cdot \frac{4095}{V_{ref}} \tag{A7}$$

K_N is the normalization gain, which is equal to the reciprocal of the bus voltage V_B, and is used to convert the control voltage from the controller output to the duty cycle. V_{ff} is the feedforward voltage. V_o is the output voltage of the converter. When operating in buck mode, it corresponds to the SC voltage, and when operating in boost mode, it corresponds to the bus voltage V_B. Simplify (A5) using (A6) and (A7).

$$d = \frac{1}{N_B} \left(N_e \cdot \frac{1}{A_v A_i R_s} G + N_{\mathrm{ff}} \right) \tag{A8}$$

The coefficient before G is the final gain of the FMAC.

$$K = \frac{1}{A_v A_i R_s} = 1 \tag{A9}$$